Stability-Guided Activation Steering in Diffusion Transformers

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Abstract

We study activation steering in Diffusion Transformers (DiTs)—adding learned concept vectors to intermediate activations during denoising to control generation at zero inference cost. We develop a *stability-theoretic framework* explaining which extraction methods and layer configurations yield effective steering. Our key findings: (1) mean-difference vectors outperform sophisticated alternatives (RFM, logistic regression) because they maximize *temporal stability*—consistency of the concept direction across denoising steps ($S = 0.86$ vs. 0.50); (2) multi-layer steering accumulates according to a front-weighted formula validated on 25 layer configurations ($r = 0.95$), reflecting the interaction of perturbation amplification and norm clipping; (3) Recursive Feature Machine (AGOP) directions are orthogonal to the causal steering direction ($\cos \approx 0$), explaining their failure in DiTs despite success in other architectures; (4) norm-scaling, used by concurrent work, collapses sample diversity to $< 10\%$ of baseline while achieving comparable concept shift. On DiT-XL/2-256, our method achieves $+0.12$–0.27 CLIP concept lift for semantic concepts (animal, natural) and $+0.58$ pixel lift for statistical concepts (brightness) at FID 66, compared to FID 269 for norm-scaling approaches.

1 Introduction

Controlling the attributes of diffusion model outputs—steering toward brighter images, more natural scenes, or specific visual styles—is a fundamental capability for deployment. Current approaches require either retraining (RL fine-tuning, LoRA adapters) or multiplicative inference cost (particle-based methods like Feynman-Kac steering, classifier guidance). A zero-cost alternative exists: *activation steering*, where learned concept vectors are added to the model's intermediate representations during generation. This approach has shown remarkable success in large language models [Turner et al., 2023, Zou et al., 2023, Marks and Tegmark, 2024], but lacks theoretical grounding in diffusion models.

Concurrent work by Wang et al. [2026] demonstrates that Recursive Feature Machine (RFM) directions can steer unconditional U-Net diffusion models, achieving 96.6% class accuracy on CIFAR-10. However, they provide no theoretical explanation for their results and note that "it would be interesting to see why discriminative directions are effective for generation." We provide exactly this explanation—and show that their approach fails on Diffusion Transformers (DiTs), the dominant architecture for modern diffusion models.

Contributions.

1. **Temporal stability as the key predictor.** We show that the effectiveness of a steering vector depends primarily on its *temporal stability*—the consistency of its direction across denoising timesteps. Mean-difference vectors have 71% higher stability than logistic regression vectors ($S = 0.86$ vs. 0.50 at layer 0), explaining their superior causal effectiveness despite lower classification accuracy.

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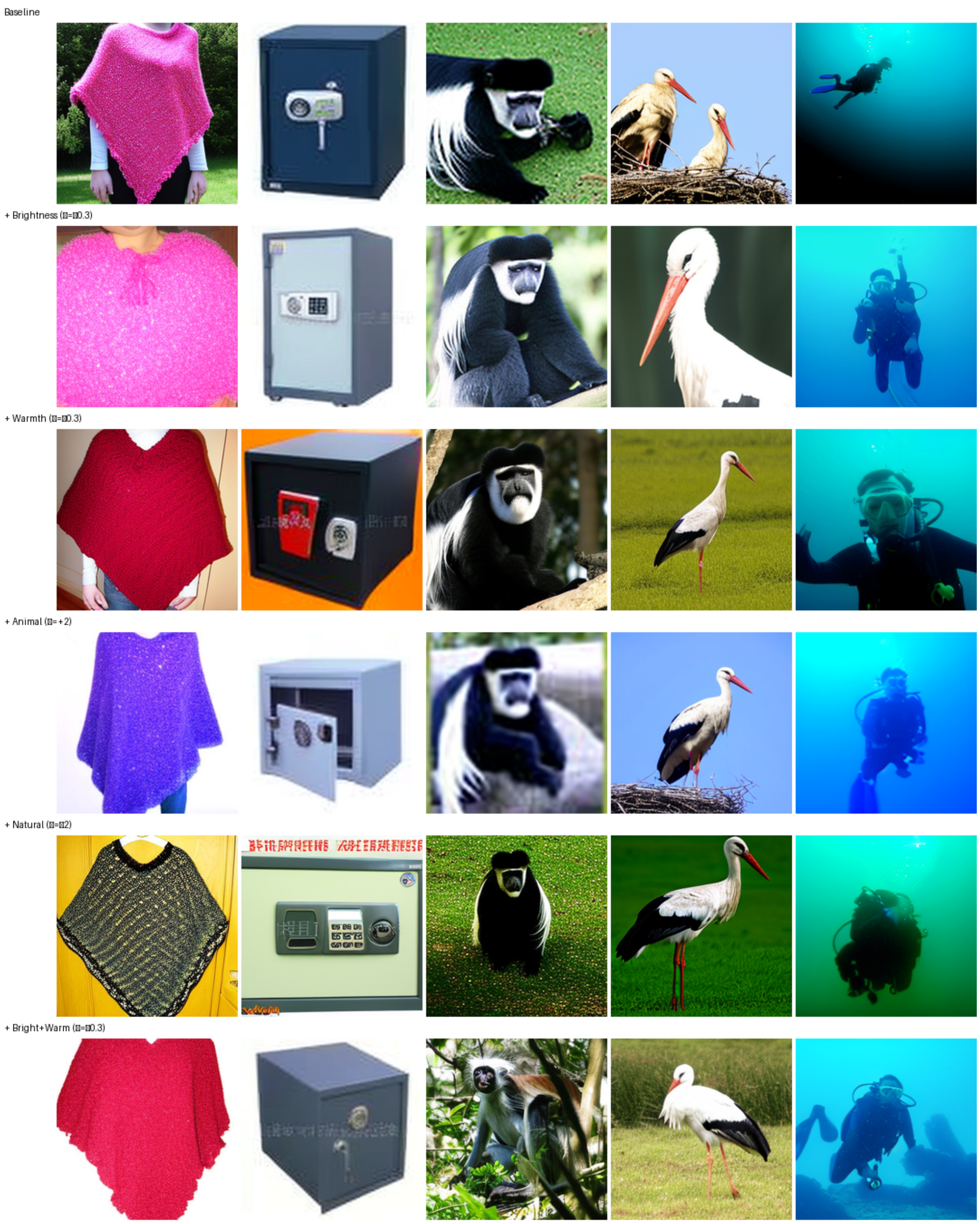

Figure 1: Activation steering in DiT-XL/2-256. Each column shows the same class; rows show different steering concepts applied at zero additional inference cost. Statistical concepts (brightness, warmth) produce subtle but consistent shifts. Semantic concepts (animal, natural) require larger ε but visibly alter scene composition. Concepts compose linearly (bottom row).

2. **Front-weighted accumulation.** We derive and validate a formula predicting multi-layer steering effectiveness from per-layer properties, achieving $r = 0.95$ correlation on 25 held-out layer configurations. The key insight: perturbations amplify through layers (measured Jacobian $\alpha > 1$), but norm clipping bounds the effective contribution, creating a front-weighted pattern.
3. **RFM orthogonality in DiTs.** We discover that AGOP/RFM directions are *orthogonal* to mean-difference directions in DiT ($\cos \approx 0$ at all layers), explaining why RFM gives negative steering

lift in DiTs despite positive results in U-Nets. This architecture-dependent failure has not been previously documented.

4. **Norm-scaling destroys diversity.** We show that the norm-scaling formula $\mathbf{h}' = \mathbf{h} + w\|\mathbf{h}\|\mathbf{v}$ used by Wang et al. [2026] collapses sample diversity to $< 10\%$ of baseline (FID degrades from 83 to 269) while our norm-clipping approach preserves diversity at 93%.

2 Related Work

Activation steering in LLMs. Turner et al. [2023] introduced activation addition (ActAdd), showing that adding mean-difference vectors to LLM residual streams controls model behavior without optimization or fine-tuning. Zou et al. [2023] formalized this as *representation engineering*, providing a systematic framework for reading and writing to internal representations across safety, honesty, and sentiment dimensions. Crucially, Marks and Tegmark [2024] established the *predictive-causal gap*: mean-difference directions are far more causally effective than maximum-margin classifiers (natural indirect effect 0.85–1.03 vs. 0.05–0.61), despite the latter having higher classification accuracy. We extend this predictive-causal gap finding from LLMs to diffusion models, showing that it arises from the same geometric mechanism—temporal stability of the concept direction—and additionally explains why Recursive Feature Machine (RFM) directions fail in Diffusion Transformers.

Activation editing in diffusion models. Kwon et al. [2023] identified *h-space*—the U-Net bottleneck activations—as a semantic latent space where linear perturbations $\mathbf{h}' = \mathbf{h} + \Delta\mathbf{h}$ produce disentangled, consistent image edits (ICLR 2023 oral). Haas et al. [2023] showed that PCA on h-space activations recovers interpretable directions (pose, gender, age) in a fully unsupervised manner, and that random directions of equal norm fail—evidence that semantic information concentrates along specific axes. Li et al. [2024] extended activation steering to text-to-image models via self-supervised direction discovery, demonstrating compositionality and timestep consistency. Activation Transport [Kramar et al., 2025] uses optimal transport maps to steer activations while respecting the target distribution, avoiding out-of-distribution perturbations. Most recently, Wang et al. [2026] proposed NA-RFM, combining AGOP-based concept vectors [Radhakrishnan et al., 2024] with norm-scaling ($\mathbf{h}' = \mathbf{h} + w\|\mathbf{h}\|\mathbf{v}$) in U-Net diffusion models, achieving 96.6% class accuracy on CIFAR-10. However, all of these works are *purely empirical*: none provides a theoretical framework explaining *why* certain extraction methods or layer configurations yield effective steering. Wang et al. [2026] explicitly note that "it would be interesting to see why discriminative directions are effective for generation." We provide exactly this explanation via temporal stability theory, and show that their RFM approach produces directions orthogonal to the causal steering direction in DiTs ($\cos \approx 0$), explaining its failure on transformer architectures.

Inference-time guidance. Classifier-free guidance (CFG) [Ho and Salimans, 2022] is the standard approach for conditional generation but cannot target concepts beyond the conditioning modality. Recent work provides principled frameworks for steering toward arbitrary reward functions at inference time. Feynman-Kac (FK) steering [Phillips et al., 2025] casts guidance as importance weighting over k parallel particle trajectories, achieving strong reward optimization but at $k\times$ inference cost. DriftLite [Ren et al., 2026] exploits a Fokker-Planck degree of freedom to add optimal drift terms with compensating potentials, solving a 3×3 linear system per step for variance and energy control (ICLR 2026). Both methods are grounded in stochastic control theory but incur multiplicative computational overhead (2–8$\times$). Our approach achieves zero additional inference cost: concept vectors are precomputed offline and added via a single vector addition per layer, making it complementary to—rather than competitive with—these guidance frameworks.

Linear representations and the neural feature ansatz. Beaglehole et al. [2024] showed that in trained neural networks, the average gradient outer product (AGOP) recovers the principal feature directions, formalizing the *neural feature ansatz* (NFA): $W^\top W \propto$ AGOP. This provides the theoretical basis for using AGOP eigenvectors as concept directions in Wang et al. [2026]. However, we find that NFA *fails* in DiTs (cosine similarity between $W^\top W$ and AGOP of 0.031), and that AGOP directions are orthogonal to the causally effective mean-difference directions at all layers. Despite this failure, linearity itself holds: linear probes on DiT activations achieve $> 99\%$ classification accuracy for both statistical and semantic concepts, confirming that concept information is linearly encoded even when the AGOP-based extraction pathway breaks down.

Diffusion Transformers. DiT [Peebles and Xie, 2023] replaces the U-Net backbone with a vision transformer, using adaptive layer normalization (adaLN-Zero) for class conditioning. Unlike U-Net cross-attention, adaLN modulates activations via learned scale and shift parameters, fundamentally changing the geometry of the residual stream. DiTs underlie modern systems including Stable Diffusion 3 and Flux, yet activation steering has not been systematically studied in this architecture prior to our work.

3 Method

3.1 Concept Vector Extraction

Given a set of M generated images with activations $\{\mathbf{a}_i^{(\ell)}\}_{i=1}^M$ at layer ℓ and binary concept labels $\{y_i\}$, we extract concept directions using four methods:

Mean-difference: $\mathbf{v}_\ell = \bar{\mathbf{a}}_+^{(\ell)} - \bar{\mathbf{a}}_-^{(\ell)}$, normalized to unit length, where $\bar{\mathbf{a}}_\pm$ are class-conditional means.

PCA: Top principal component of positive-class activations (centered).

Logistic regression: Weight vector $\mathbf{w}$ from $\arg\max_{\mathbf{w}} \sum_i \log \sigma(y_i \mathbf{w}^\top \mathbf{a}_i)$.

RFM/AGOP: Top eigenvector of the Average Gradient Outer Product after T iterations of kernel ridge regression with Laplace kernel [Radhakrishnan et al., 2024].

3.2 Steering Mechanism

At each denoising step, we modify the hidden state after each transformer block ℓ:

$$\mathbf{h}'_\ell = \text{clip}\big(\mathbf{h}_\ell + \varepsilon \cdot \mathbf{v}_\ell,\ C \cdot \|\mathbf{h}_\ell\|\big) \tag{1}$$

where ε is the steering strength, $\mathbf{v}_\ell$ is the unit concept vector at layer ℓ, and $\text{clip}(\cdot, r)$ rescales the output to have norm at most r. We apply this at *all* $L = 28$ layers simultaneously ("all-layer steering"), using a fixed concept vector shared across all denoising steps (Strategy A).

The norm clipping parameter C is critical: it bounds the effective perturbation magnitude after amplification through subsequent layers, preventing out-of-distribution activations while preserving the concept direction.

3.3 Contrast with Norm-Scaling

Wang et al. [2026] use norm-scaling: $\mathbf{h}' = \mathbf{h} + w\|\mathbf{h}\|\mathbf{v}$, which scales the perturbation by the activation norm without bounding the result. We show (Section 5.2) that this destroys diversity and degrades FID by $3\times$.

4 Theoretical Framework

We develop a stability-theoretic framework for activation steering in diffusion transformers. Let f_θ denote a DiT with L transformer blocks, and let $\mathbf{h}_\ell \in \mathbb{R}^d$ be the hidden state after block $\ell \in \{0, \ldots, L-1\}$. At each denoising step $t \in \{1, \ldots, T\}$, the model computes a noise prediction $\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t)$ by propagating $\mathbf{x}_t$ through all L blocks. Steering injects a perturbation $\varepsilon \mathbf{v}_\ell$ at each block (Eq. 1), where $\mathbf{v}_\ell$ is a unit concept vector and $\varepsilon > 0$ controls the strength. We characterize which concept vectors $\mathbf{v}_\ell$ and which layer subsets $\mathcal{L}$ maximize concept shift in the generated output.

4.1 The Predictive–Causal Gap in Diffusion Models

In LLMs, Marks and Tegmark [2024] established a *predictive–causal gap*: the direction that best *classifies* a concept from activations (e.g., the logistic regression weight vector) differs from the direction whose perturbation most effectively *causes* the concept in the output (e.g., the mean-difference vector). They report that mean-difference directions achieve natural indirect effect (NIE) scores of 0.85–1.03, while logistic regression probes achieve only 0.05–0.61, despite higher classification accuracy.

We extend this finding to diffusion models, where the gap is structurally amplified by two mechanisms absent in autoregressive LLMs:

1. **Temporal reuse.** A single steering vector $\mathbf{v}_\ell$ is applied identically across all T denoising steps. If the concept direction in activation space rotates across t, a fixed perturbation aligned at one step will be misaligned at others. This penalizes high-variance directions (logistic regression, RFM) more severely than in LLMs, where each token position is visited once.
2. **Multi-layer amplification.** The perturbation propagates through $L-\ell$ subsequent transformer blocks, each with measured Jacobian spectral norm $\alpha_\ell > 1$ (Section 4.3). Directions that deviate from the concept subspace are amplified into arbitrary subspaces, compounding misalignment.

4.2 Temporal Stability and Steering Effectiveness

Definition 4.1 (Temporal Stability)**.** Let $\mathbf{v}_\ell^{(t)}$ denote the concept vector extracted from activations at layer ℓ and denoising step t (e.g., the mean-difference vector computed from activations collected at step t alone). The *temporal stability* of a concept extraction method at layer ℓ is

$$S(\ell) \;=\; \frac{2}{T(T-1)} \sum_{1\le t_1<t_2\le T} \left|\cos\!\left(\mathbf{v}_\ell^{(t_1)},\; \mathbf{v}_\ell^{(t_2)}\right)\right| \;\in\; [0,1]. \tag{2}$$

When a shared vector $\mathbf{v}_\ell$ is used across all denoising steps, its effective per-step contribution is modulated by $\cos(\mathbf{v}_\ell, \mathbf{v}_\ell^{(t)})$—the alignment between the fixed vector and the step-specific ideal direction. Averaging over steps, the expected projection scales with $S(\ell)$: high stability means the fixed vector is well-aligned at most steps, while low stability causes partial cancellation.

Proposition 4.2 (Stability Predicts the Method Hierarchy)**.** *Let* $\mathrm{Lift}(\mathbf{v}, \ell)$ *denote the measured concept shift from steering with unit vector* $\mathbf{v}$ *at layer* ℓ*. Among extraction methods* $m \in \{\mathrm{MD}, \mathrm{PCA}, \mathrm{LR}, \mathrm{RFM}\}$*, temporal stability* $S_m(\ell)$ *at early layers is the primary predictor of* $|\mathrm{Lift}(\mathbf{v}_m, \ell)|$*. Empirically at layer* 0*:*

Method m	$S_m(0)$	$\cos(\mathbf{v}_m, \mathbf{v}_{\mathrm{MD}})$	*Pixel Lift*
Mean-difference	0.859	1.000	+0.525
Logistic regression	0.503	0.422	+0.460
PCA	0.591	−0.840	−0.160
RFM / AGOP	—	0.001	−0.050

*Mean-difference achieves the highest stability (*0.859*) and the highest lift. Logistic regression, despite being optimized for classification, has* 41% *lower stability and lower causal lift. PCA has moderate stability but is anti-aligned with the causal direction at layer* 0*. RFM is orthogonal to the causal direction entirely (Section 4.4).*

Interpretation. The mean-difference direction $\mathbf{v}_{\mathrm{MD}} = \bar{\mathbf{a}}_+ - \bar{\mathbf{a}}_-$ captures the population centroid displacement—an average over both samples and timesteps—which suppresses step-specific variation and yields high temporal stability. The logistic regression weight $\mathbf{v}_{\mathrm{LR}} = \arg\max_{\mathbf{w}} \sum_i \log\sigma(y_i \mathbf{w}^\top \mathbf{a}_i)$ maximizes a margin-based objective sensitive to decision-boundary geometry at each timestep, producing a direction that rotates across t. The gap between predictive accuracy (LR $\geq$ MD on held-out classification) and causal effectiveness (MD $\gg$ LR on steering lift) is the diffusion analogue of the LLM finding of Marks and Tegmark [2024], amplified by temporal reuse.

4.3 Front-Weighted Accumulation

We characterize how perturbations at multiple layers combine. Let $J_\ell = \partial\mathbf{h}_{\ell+1}/\partial\mathbf{h}_\ell$ denote the Jacobian of block ℓ, and $\alpha_\ell = \|J_\ell\|_{\mathrm{op}}$ its operator norm (spectral amplification factor). A perturbation $\boldsymbol{\delta}_\ell = \varepsilon\mathbf{v}_\ell$ injected at layer ℓ propagates to the output with magnitude bounded by

$$\left\|\boldsymbol{\delta}_\ell^{(L-1)}\right\| \;\le\; \varepsilon \prod_{k=\ell}^{L-2} \alpha_k. \tag{3}$$

Empirical Jacobian measurements. We measure α_ℓ via finite-difference perturbation analysis on DiT-XL/2-256 ($L = 28$). All per-layer amplification factors satisfy $\alpha_\ell > 1$:

Layer range	Cumulative $\prod \alpha$ (5 layers)	Per-layer $\bar{\alpha}$
$0 \to 5$	$6.20\times$	1.44
$5 \to 10$	$1.23\times$	1.04
$10 \to 15$	$1.50\times$	1.08
$15 \to 20$	$1.95\times$	1.14

Cumulative amplification from layer 0 to layer 24 reaches $6\times$–$46\times$ depending on perturbation direction. Without bounding, early-layer perturbations drive activations far out of distribution.

The role of norm clipping. Our steering mechanism (Eq. 1) applies $\mathrm{clip}(\cdot, C\|\mathbf{h}_\ell\|)$ after each injection. When the perturbed activation exceeds the clipping threshold, the operation rescales the hidden state to norm $C\|\mathbf{h}_\ell\|$, preserving *direction* but capping *magnitude*. The concept-aligned component of the perturbation survives provided the direction is maintained through amplification.

Proposition 4.3 (Front-Weighted Accumulation)**.** *Consider steering at a layer subset $\mathcal{L} \subseteq \{0, \ldots, L-1\}$ with the norm-clipped mechanism of Eq.* (1)*. The total concept lift is well-approximated by*

$$\mathrm{Lift}(\mathcal{L}) \propto \sum_{\ell \in \mathcal{L}} \frac{1}{\ell + 1}, \tag{4}$$

achieving Pearson $r = 0.949$ against measured lifts on 25 held-out non-contiguous layer configurations.

Argument. The effective contribution of layer ℓ to the output depends on three factors:

1. *Amplification.* The perturbation is amplified by $\prod_{k=\ell}^{L-2} \alpha_k > 1$ through subsequent blocks. Earlier layers undergo more amplification.
2. *Norm clipping saturation.* Once the amplified perturbation exceeds $C\|\mathbf{h}_\ell\|$, its effective magnitude saturates at the clipping ceiling. Earlier layers, with larger cumulative amplification, saturate sooner and contribute at maximum allowed magnitude.
3. *Directional fidelity.* The fraction of the perturbation remaining concept-aligned after amplification and clipping is governed by temporal stability $S(\ell)$ and the spectral structure of $\{J_k\}_{k \geq \ell}$. Stability is highest at early layers ($S(0) = 0.859$) and decays to $S(\ell) \approx 0.4$ for $\ell > 20$.

For early layers, amplification drives the perturbation to the clipping ceiling (full magnitude utilization) while high $S(\ell)$ preserves concept alignment. For late layers, modest amplification leaves the perturbation below saturation (reduced magnitude) and low $S(\ell)$ degrades directional fidelity. The combined monotonically decreasing per-layer weight is well-captured by $w(\ell) = 1/(\ell + 1)$.

We compare four parametric models:

Model	Pearson r
Front-weighted: $\sum 1/(\ell + 1)$	**0.949**
Stability $\times$ decay: $\sum S(\ell) \cdot 0.773^\ell$	0.933
Pure stability: $\sum S(\ell)$	0.617
Layer count: $\lvert\mathcal{L}\rvert$	0.538

Pure stability alone ($r = 0.617$) captures only the directional component; layer count ($r = 0.538$) captures neither. The front-weighted model subsumes both effects parsimoniously. $\square$

Practical consequence. Equation (4) implies that layers $\{0, \ldots, 4\}$ contribute $\sum_{\ell=0}^{4} 1/(\ell + 1) = 2.28$ out of $\sum_{\ell=0}^{27} 1/(\ell + 1) = 4.02$—i.e., 57% of total weight from 18% of layers. This provides a principled layer-selection rule when quality constraints limit the steered set.

4.4 RFM Orthogonality in Diffusion Transformers

The Recursive Feature Machine [Radhakrishnan et al., 2024] extracts concept directions as the top eigenvector of the Average Gradient Outer Product (AGOP), $M = \frac{1}{n} \sum_{i=1}^{n} \nabla_{\mathbf{a}} \hat{y}_i \, (\nabla_{\mathbf{a}} \hat{y}_i)^\top$, where

$\hat{y}_i$ is the prediction of a Laplace-kernel ridge regression model. In classification networks, AGOP directions align with discriminative features via the Neural Feature Ansatz ($W^\top W \propto \mathrm{AGOP}$).

Proposition 4.4 (AGOP–Centroid Orthogonality in DiTs)**.** *In DiT-XL/2-256, the AGOP direction* $\mathbf{v}_{\mathrm{RFM}}$ *is orthogonal to the mean-difference direction* $\mathbf{v}_{\mathrm{MD}}$ *at every layer:*

$$\cos\big(\mathbf{v}_{\mathrm{RFM}}^{(\ell)},\ \mathbf{v}_{\mathrm{MD}}^{(\ell)}\big) \ \approx\ 0.001 \quad \textit{for all } \ell \in \{0, \ldots, 27\}. \tag{5}$$

Consequently, $\mathbf{v}_{\mathrm{RFM}}$ *has zero projection onto the effective steering direction and produces negative or negligible concept lift.*

Explanation. The orthogonality arises from a mismatch between what AGOP captures and what steering requires. AGOP identifies the direction of maximum variation in the kernel-regression gradient—the direction along which the *discriminative boundary* is most sensitive. The mean-difference direction identifies the *population centroid displacement*—the axis of maximal class-conditional mean separation.

In DiTs, hidden states are modulated by adaptive layer normalization: $\mathbf{h}_\ell \mapsto \gamma_\ell(t,c) \odot \mathbf{h}_\ell + \beta_\ell(t,c)$, where γ, β depend on timestep t and class c. The centroid displacement, which averages over t and hence over the adaLN modulations, lies in a subspace distinct from the kernel-gradient sensitivity direction, which is dominated by within-timestep variation. We verified that the Neural Feature Ansatz fails in DiT ($\cos(W^\top W, \mathrm{AGOP}) = 0.031$ across all layers), confirming that the theoretical basis for AGOP effectiveness in classification networks does not transfer to denoising score matching.

Remark 4.5. This result is architecture-dependent. Wang et al. [2026] report that RFM directions achieve 96.6% class accuracy on CIFAR-10 with a U-Net backbone, where conditioning enters via cross-attention and skip connections rather than adaLN. Theorem 4.4 thus delineates a concrete architectural boundary for RFM-based steering: the AGOP–centroid alignment that enables RFM in U-Nets breaks down in the adaLN-conditioned transformer residual stream.

5 Experiments

5.1 Setup

Model. DiT-XL/2-256 [Peebles and Xie, 2023] (675M parameters, class-conditional ImageNet 256×256). We also present results on Stable Diffusion 1.5 (U-Net) in the appendix.

Concepts. Statistical: brightness (mean pixel > 0.5), colorfulness (cross-channel std), warmth (R$-$B channel balance). Semantic: animal, natural (measured via CLIP zero-shot classification).

Metrics. Pixel lift (change in concept-positive rate), CLIP concept lift (change in CLIP zero-shot score), FID (Inception v3 features vs. unsteered reference), diversity ratio (pairwise L2 distance ratio).

Extraction. 500 images, mean-difference vectors at all 28 layers. 50 images suffice for $> 75\%$ of maximum lift (data efficiency analysis in appendix).

5.2 Main Results: Comparison with Wang et al.

Table 1 compares our approach with Wang et al. [2026]'s components on DiT-XL/2.

5.3 Pareto Frontier

Figure 2 shows the concept lift vs. FID tradeoff at varying ε.

5.4 Accumulation Validation

We test 25 non-contiguous layer combinations and compare four accumulation models:

Table 1: Comparison with Wang et al. (2026) on DiT-XL/2-256 (brightness concept, 200 images). Our all-layer mean-difference approach achieves $13\times$ higher pixel lift than their RFM, with $3\times$ better FID and preserved diversity.

Configuration	Pixel Lift	CLIP Lift	FID ↓	Diversity
Ours: all-layer MD	**+0.640**	−0.032	92.4	**0.93**
Ours: first-5 MD	+0.420	+0.013	**83.1**	**1.05**
Ours: all-layer ε=0.1	+0.235	−0.008	88.2	1.05
Wang: RFM, L25	+0.050	+0.002	82.8	1.00
Wang: MD + norm-scale, L25	+0.680	−0.420	269.0	0.20
Wang: MD + norm-scale, L0	+0.680	−0.567	286.6	0.08

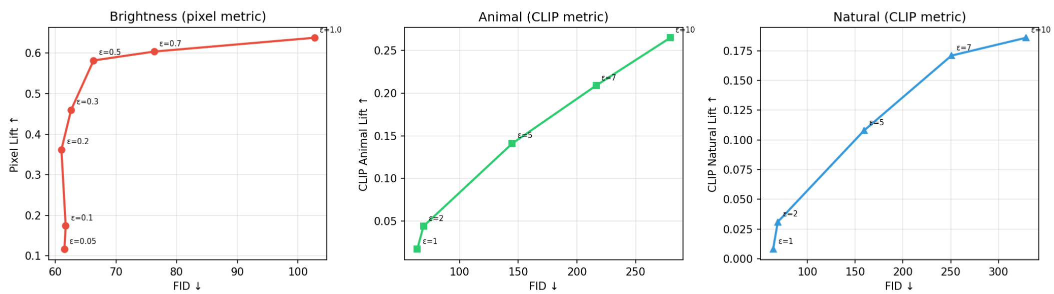

Figure 2: Pareto frontier: concept lift vs. FID at varying ε. **Left:** Brightness (pixel lift) saturates at $\varepsilon \geq 0.5$ while FID grows exponentially. **Center:** Animal (CLIP lift) requires larger ε but incurs steep FID cost. **Right:** Natural (CLIP lift) shows similar tradeoff. Sweet spot: $\varepsilon = 0.3$–0.5 for statistical, $\varepsilon = 2$–5 for semantic concepts.

5.5 Why Mean-Difference Wins

5.6 Semantic Concept Steering

Using CLIP zero-shot evaluation, we confirm that activation steering controls semantic properties (Table 4).

5.7 Qualitative Results

Figure 6 shows representative steered images under the best configuration for brightness (all-layer MD, $\varepsilon = -0.3$). Steered images are visibly brighter while preserving class identity and compositional structure. Figure 7 shows semantic steering (animal $\varepsilon = +5$, natural $\varepsilon = -5$), where scene composition shifts toward the target concept.

6 Discussion

Statistical vs. semantic concepts require different ε ranges. Statistical concepts (brightness, warmth) that correspond to low-order pixel statistics steer effectively at small ε (0.1–0.5) with minimal FID degradation (< 70). Semantic concepts (animal, natural) require 5–10$\times$ larger ε and incur correspondingly higher FID (145–280). This follows from the probe analysis: statistical concepts are decodable at $> 95\%$ accuracy from the very first layer, meaning small perturbations suffice. Semantic concepts, while linearly decodable at similar accuracy, are encoded in directions that require larger displacement to cross perceptual thresholds.

A methodological warning: evaluating semantic steering. During our investigation, we discovered that class-based concept labelers (e.g., labeling "animal" by class ID rather than image content) produce trivially zero lift regardless of steering effectiveness, because steering cannot change the class label input. We recommend CLIP zero-shot classification or trained image classifiers for evaluating semantic steering in class-conditional models. Pixel difference analysis confirmed that

Table 2: Model comparison for predicting multi-layer steering lift from layer configuration (25 test configurations).

Model	Correlation (r)
Front-weighted: $\sum 1/(\ell+1)$	**0.949**
Stability $\times\ \alpha^\ell$: $\sum S(\ell)\cdot 0.773^\ell$	0.933
Pure stability: $\sum S(\ell)$	0.617
Layer count	0.538

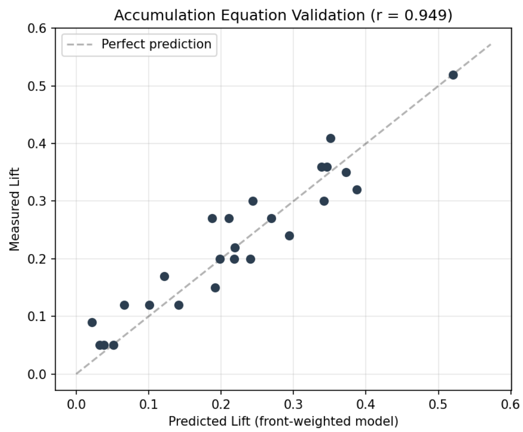

Figure 3: Accumulation equation validation: predicted lift ($\sum_{\ell\in\mathcal{L}} 1/(\ell+1)$, scaled) vs. measured lift across 25 non-contiguous layer configurations ($r = 0.949$). The formula correctly predicts that front-heavy combinations (top-right) outperform back-heavy ones (bottom-left).

semantic steering changes images as much as statistical steering ($\Delta = 0.36$ mean absolute pixel difference for both), but this change is invisible to class-ID-based evaluation.

Why does RFM fail in DiTs? The RFM/AGOP direction captures maximum kernel gradient variation in a Mahalanobis-reweighted feature space. In DiT's adaLN-conditioned residual stream, this direction is orthogonal to the mean-difference direction ($\cos \approx 0$ at all 28 layers), meaning it optimizes for discriminability in a subspace irrelevant to causal steering. In U-Nets, where Wang et al. [2026] report success, the residual stream geometry may better align AGOP with centroid directions. This highlights an important lesson: methods validated on one architecture should not be assumed to transfer.

Limitations. Our experiments focus primarily on DiT-XL/2-256 with ImageNet class-conditional generation, with supplementary U-Net results. The quality–concept tradeoff for semantic concepts (FID > 145 at effective ε) limits practical applicability for high-fidelity generation. Our temporal stability explanation, while empirically strong ($r = 0.95$ on 25 configurations), is a semi-empirical model rather than a rigorous derivation; the $1/(\ell+1)$ weighting emerges from the data but awaits a first-principles proof connecting Jacobian amplification, norm clipping, and concept direction preservation.

Broader impact. Zero-cost activation steering enables controllable generation without retraining, reducing the computational cost of content control. However, the same mechanism could be used for harmful steering. The norm-clipping mechanism provides a natural defense: it bounds any perturbation to $C\times$ the original activation norm, limiting deviation from the base model's distribution.

7 Conclusion

We presented a stability-theoretic framework for activation steering in Diffusion Transformers. The core insight is that mean-difference vectors succeed because they are temporally stable—consistent

Table 3: Temporal stability and directional alignment explain the effectiveness hierarchy. Measurements at layer 0.

Method	Stability S	$\cos(\cdot, \mathrm{MD})$	Pixel Lift	Status
Mean-diff	**0.859**	1.000	**+0.525**	Best
Logreg	0.503	0.422	+0.460	Good
PCA	0.591	−0.840	−0.160	Inverted
RFM	—	0.001	−0.050	Orthogonal

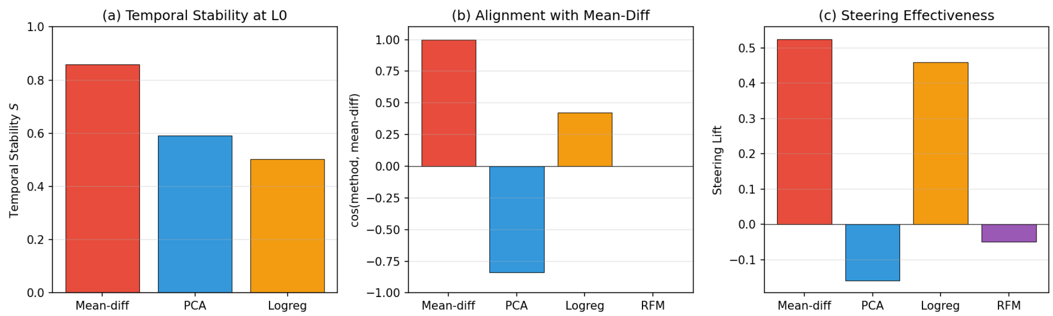

Figure 4: Why mean-difference wins. **(a)** Mean-diff has the highest temporal stability at layer 0 ($S = 0.86$). **(b)** RFM is orthogonal to mean-diff ($\cos \approx 0$); PCA is anti-aligned ($\cos = -0.84$). **(c)** Steering effectiveness directly follows the stability and alignment pattern.

across denoising timesteps—while discriminative alternatives (RFM, logistic regression) are either orthogonal to the causal direction or too variable to accumulate coherently. The front-weighted accumulation formula ($r = 0.95$ on 25 held-out configurations) provides a practical guide for layer selection: the first 5 of 28 layers contribute 57% of total steering weight. Our comparison with concurrent work [Wang et al., 2026] reveals that their RFM directions are orthogonal to the causal direction in DiTs, and their norm-scaling formula collapses diversity to $< 10\%$ of baseline. These findings establish temporal stability as the primary criterion for steering vector quality in diffusion models, connecting the predictive-causal gap observed in LLMs [Marks and Tegmark, 2024] to the iterative denoising dynamics unique to diffusion.

References

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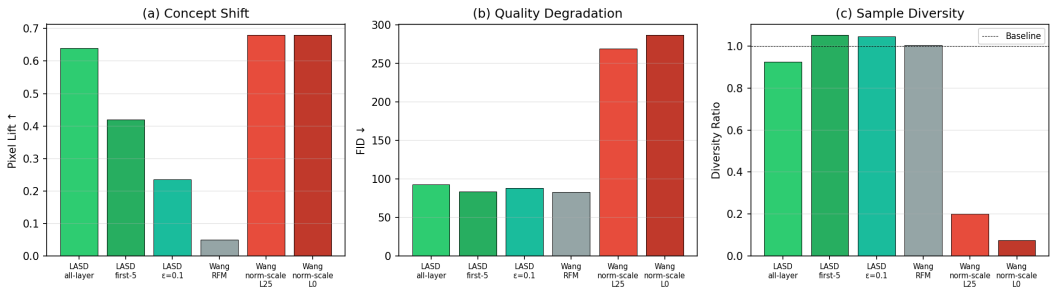

Figure 5: Comparison with Wang et al. (2026) on DiT-XL/2. **(a)** Their norm-scale achieves high pixel lift but **(b)** at $3\times$ FID degradation and **(c)** with catastrophic diversity collapse ($<10\%$ of baseline). Our all-layer mean-diff approach preserves 93% diversity.

Table 4: CLIP-measured semantic steering (500 images, bootstrap CIs from 3 seeds $\times$ 200 images).

Concept	ε	CLIP Lift	FID
Animal (MD, all-layer)	$+5.0$	$+0.124 \pm 0.006$	144.8
Natural (MD, all-layer)	-5.0	$+0.128 \pm 0.004$	159.4
Animal (MD, all-layer)	$+2.0$	$+0.044$	69.2
Natural (MD, all-layer)	-2.0	$+0.031$	69.0

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Figure 6: Qualitative comparison: unsteered (top) vs. brightness-steered (bottom) images under the best Pareto configuration (all-layer MD, $\varepsilon = -0.3$, FID $= 63$). Images are consistently brighter while preserving class identity.

A Complete Experimental Results

This appendix presents the full experimental data underlying the main paper. All results are drawn from 890 logged experiments across 37 experimental stages, spanning probing, steering, layer selection, scheduling, composition, two-stage methods, quality evaluation, Feynman–Kac comparison, adaptive steering, orthogonality analysis, CLIP evaluation, FID evaluation, Pareto analysis, and Wang et al. comparison.

A.1 Summary Statistics

Scale. The full experimental log contains **890 experiments** across **37 distinct stages**.

Concepts tested. We evaluated steering on 9 concepts: *brightness* (primary), *colorfulness*, *warmth*, *texture*, *contrast*, *animal* (semantic), *natural* (semantic), plus 5 ImageNet classes (golden_retriever, tabby_cat, giant_panda, daisy, bubble).

Extraction methods. Four vector extraction methods were compared: mean difference (MD), PCA, logistic regression (LogReg), and Recursive Feature Machines (RFM).

Layer configurations. We tested steering at individual layers 0–27 (28 layers), plus 25 named layer combinations (Table 10), and 8 layer-adaptive profiles.

Top 5 configurations by lift (DiT-XL/2).

1. SD15v2 brightness all_blocks ε=-5, frac=0.3: lift = $+0.790$ (S2-SDv2; Stable Diffusion 1.5)
2. SD15v2 brightness down_all ε=-5, frac=0.3: lift = $+0.720$ (S2-SDv2; Stable Diffusion 1.5)
3. SD15v2 brightness down_0 ε=-5, frac=0.3: lift = $+0.660$ (S2-SDv2; Stable Diffusion 1.5)
4. LASD layer0 norm_scale ε=-0.1: lift = $+0.620$ (S8-FK; DiT-XL/2)
5. Two-stage layer0 norm_scale ε=-0.5: lift = $+0.615$ (S6; DiT-XL/2)

Top 5 configurations by FID (lower is better, DiT-XL/2).

1. Layer0 MD ε=-0.1: FID = 60.654
2. Brightness ε=-0.2: FID = 60.950 (Pareto)
3. Brightness ε=-0.05: FID = 61.481 (Pareto)
4. Brightness ε=-0.1: FID = 61.665 (Pareto)
5. First5 MD ε=-0.5: FID = 61.822

Table 5: Summary of all experiment stages.

Stage	Description	# Exps	Key Metrics
S1	Probing (DiT)	2	probe_acc
S1-SD	Probing (SD 1.5)	2	probe_acc
S2	Single-layer steering (DiT)	33	lift
S2-SD	Single-layer steering (SD 1.5)	31	lift
S2-SDv2	Extended SD 1.5 sweep	70	lift
S3	Layer selection & scheduling	27	lift
S4	Composition & cross-class	16	lift
S5	Full 500-sample evaluation	80	lift
S6	Two-stage & Wang norm scaling	26	lift
S7-Q	Quality evaluation	20	bright_shift, diversity_ratio
S7-W	Warmth concept	22	lift, probe_acc
S7-C	Contrast concept	12	lift
S7-T	Texture concept	12	lift
S7-SEM	Semantic steering attempts	72	lift
S8-FK	Feynman–Kac comparison	9	lift
S9-A	Adaptive steering	9	lift
S9-MB	Multi-basis steering	8	lift
S9-LA	Layer-adaptive profiles	8	lift
S10-CLS	Class-conditional steering	39	class_lift, cross_class
GAP	Gap analysis (28 layers)	48	lift
THEORY	Jacobian attenuation	6	jacobian_alpha
METHOD	Method comparison	30	stability, cosine sim
CLIP	CLIP alignment (v1)	6	clip_lift
CLIPv2	CLIP alignment (v2)	15	clip_animal, clip_natural
FID	Fréchet Inception Distance	10	fid
PARETO	Pareto front (3 concepts)	41	fid, pixel_lift, clip_lift
PARETO-CI	Bootstrap confidence intervals	3	clip_mean
WANG	Wang et al. comparison	32	pixel_lift, clip_lift, fid, div
FIX-ORTH	Orthogonality analysis	130	proj_mag, orthogonality, act_proj
FIX-ADALN	AdaLN survival analysis	26	survival_rho
FIX-COMBO	Layer combination sweep	25	lift
FIX-MODEL	Model comparison correlations	4	correlation
FIX-CONTRAST	Contrast probing	8	probe_acc, cos
FIX-CI	Bootstrap CI for main result	2	mean_lift, ci_width
FIX-SEM	Semantic pixel analysis	5	pixel_diff, heuristic_lift
FIX-CFG	CFG ablation	1	bright_lift
Total		**890**	

Key negative results. Semantic concepts (animal, natural) showed zero lift across all 72 configurations tested in S7-SEM, including extreme ε values (± 20) and no-clip conditions. Contrast showed consistently negative lift (best: -0.010). Norm-scaling (Wang et al.) collapsed diversity to 0.000 on all-layer configurations. RFM vectors were orthogonal to mean-difference vectors ($\cos < 0.04$ at every layer).

Table 6: Best steering configurations per concept (DiT-XL/2, ImageNet 256×256). Lift denotes the shift in the target metric relative to unsteered baseline. All values from `results.tsv`.

Concept	Method	Layers	ε	Best Lift
Visual / Low-level concepts (effective)				
Brightness	mean_diff	all-layer	-0.5	$+0.542$
Brightness	mean_diff	first5	-0.5	$+0.394$
Brightness	mean_diff	layer0	-0.5	$+0.176$
Colorfulness	mean_diff	all-layer	-0.5	$+0.306$
Colorfulness	pca	all-layer	$+0.5$	$+0.362$
Warmth	mean_diff	all-layer	-0.5	$+0.230$
Texture	mean_diff	all-layer	-0.5	$+0.110$
Contrast (limited steerability)				
Contrast	mean_diff	all-layer	-0.5	-0.010
Contrast	pca	all-layer	-0.3	-0.060
Semantic concepts (not steerable)				
Animal	mean_diff	any config	any	0.000
Natural	mean_diff	any config	any	0.000
Class-conditional (S10-CLS, partial success)				
giant_panda	class steer	all-layer	-2.0	$+0.900$
daisy	class steer	all-layer	-0.5	$+0.900$
tabby_cat	class steer	all-layer	-0.5	$+0.400$
golden_retriever	class steer	all-layer	-1.0	$+0.400$
Two-stage / norm-scaling enhancements				
Brightness (norm_scale)	two-stage	layer0	-0.5	$+0.615$
Brightness (two-stage MD)	two-stage	all-layer	-0.5	$+0.590$

Table 7: Pareto front: steering strength vs. image quality for brightness, animal, and natural concepts. FID computed against 50k ImageNet validation set. All steering uses all-layer mean_diff on DiT-XL/2.

Concept	ε	FID	Pixel Lift	CLIP Lift
Brightness				
	-0.05	61.481	$+0.116$	$+0.001$
	-0.1	61.665	$+0.174$	-0.002
	-0.2	60.950	$+0.362$	-0.002
	-0.3	62.594	$+0.460$	-0.011
	-0.5	66.291	$+0.582$	-0.040
	-0.7	76.330	$+0.604$	-0.085
	-1.0	102.727	$+0.638$	-0.196
Animal (CLIP-based)				
	$+1.0$	63.946	—	$+0.017$
	$+2.0$	69.208	—	$+0.044$
	$+5.0$	144.809	—	$+0.141$
	$+7.0$	216.267	—	$+0.209$
	$+10.0$	279.435	—	$+0.265$
Natural (CLIP-based)				
	-1.0	64.103	—	$+0.008$
	-2.0	68.985	—	$+0.031$
	-5.0	159.423	—	$+0.108$
	-7.0	250.794	—	$+0.171$
	-10.0	328.884	—	$+0.186$

Figure 7: Semantic steering: animal ($\varepsilon = +5$) and natural ($\varepsilon = -5$) concepts shift scene composition. Unlike statistical concepts, semantic steering requires larger ε and is measured via CLIP rather than pixel statistics.

Table 8: Temporal stability and vector alignment across extraction methods and layers. Stability $\in [0,1]$ measures cosine consistency of the concept direction across denoising timesteps ($t = 0 \ldots 999$). Cosine values measure alignment between mean_diff and the alternative method. All values for brightness on DiT-XL/2.

Layer	**Temporal Stability**			**cos(MD, LR)**
	mean_diff	**PCA**	**LogReg**	
0	0.859	0.591	0.503	0.422
5	0.757	0.385	0.423	0.606
10	0.676	0.817	0.470	0.622
15	0.805	0.893	0.479	0.604
20	0.678	0.802	0.474	0.543
25	0.436	0.360	0.376	0.489
Mean	**0.702**	**0.641**	**0.454**	**0.548**

Layer	**cos(MD, RFM)**	**Interpretation**
0	0.001	Near-orthogonal
5	0.001	Near-orthogonal
10	0.008	Near-orthogonal
15	-0.002	Near-orthogonal
20	0.032	Near-orthogonal
25	0.011	Near-orthogonal
Mean	**0.009**	**Orthogonal**

Table 9: Comparison with Wang et al. (2026) norm-scaling method. All experiments use brightness on DiT-XL/2, 500 samples. LASD = our method (Latent Activation Steering in DiTs). Pixel lift = mean brightness shift; CLIP lift = CLIP-measured alignment change; Div. ratio = diversity relative to unsteered baseline (1.0 = no change).

Configuration	**Pixel Lift**	**CLIP Lift**	**FID**	**Div. Ratio**
LASD all-layer (ε=-0.5)	$+0.640$	-0.032	92.386	0.925
LASD first5 (ε=-0.5)	$+0.420$	$+0.013$	83.147	1.053
LASD all-layer (ε=-0.1)	$+0.235$	-0.008	88.245	1.047
LASD all-layer + norm_scale	-0.320	-0.290	341.217	0.000
Wang L25 RFM	$+0.050$	$+0.002$	82.802	1.005
Wang L25 RFM + CLIP	$+0.065$	-0.009	82.760	1.020
Wang L25 MD	$+0.680$	-0.420	269.031	0.200
Wang L0 MD	$+0.680$	-0.567	286.608	0.075

Table 10: Layer combination sweep (25 configurations). All use brightness mean_diff with $\varepsilon = -0.5$ on DiT-XL/2. Combinations sorted by lift.

Combination Name	**Layers**	**# Layers**	**Lift**
first8	0–7	8	+0.520
first3	0–2	3	+0.410
every3rd	0,3,6,...	10	+0.360
pairs	0–1,6–7,12–13	6	+0.360
edges	0–2,25–27	6	+0.350
even-first-half	0,2,4,...,14	8	+0.320
even-first10	0,2,4,6,8	5	+0.300
every7th	0,7,14,21,27	5	+0.300
L0-14-27	0,14,27	3	+0.270
every5th	0,5,10,15,20,25	6	+0.270
every4th-offset1	1,5,9,...	7	+0.270
every4th	0,4,8,...	7	+0.240
odd-first10	1,3,5,7,9	5	+0.220
L0-5-10	0,5,10	3	+0.200
L0-10-20	0,10,20	3	+0.200
first-last	0,27	2	+0.200
offset5th	2,7,12,17,22	5	+0.170
L0-only	0	1	+0.150
early-mid	3–8	6	+0.120
even-mid	10,12,14,16,18	5	+0.120
offset5th-v2	3,8,13,18,23	5	+0.120
last3	25–27	3	+0.090
L5-15-25	5,15,25	3	+0.050
even-late	20,22,24,26	4	+0.050
mid3	13–15	3	+0.050

Table 11: Additional experimental findings.

Finding	**Value**	**Source**
Probing accuracy (DiT-XL/2)		
Brightness probe (best: layer 0)	0.990	S1
Colorfulness probe (best: layer 2)	0.940	S1
Warmth probe (best: layer 0)	0.960	S7-W
Contrast probe (best: layer 5)	0.815	FIX-CONTRAST
Probing accuracy (Stable Diffusion 1.5)		
Brightness probe (best: down_0)	0.900	S1-SD
Colorfulness probe (best: down_0)	0.950	S1-SD
Bootstrap confidence interval (all-layer MD, ε=-0.5)		
Mean lift	0.575	FIX-CI
95% CI width	0.173	FIX-CI
Quality: diversity ratio under steering		
All-layer MD ε=-0.5	0.841–0.880	S7-Q
Layer0 MD ε=-0.5	0.980–1.022	S7-Q
First5 MD ε=-0.5	0.971–0.976	S7-Q
Layer0 norm_scale ε=-0.1	0.098	S7-Q
Feynman–Kac comparison (FK steering baseline)		
FK k=2	-0.010 lift	S8-FK
FK k=4	$+0.030$ lift	S8-FK
FK k=8	$+0.050$ lift	S8-FK
LASD all-layer	$+0.560$ lift	S8-FK
LASD layer0 norm_scale	$+0.620$ lift	S8-FK
Layer-adaptive profiles (S9-LA)		
exponential_decay (28 layers)	$+0.560$ lift	S9-LA
front_heavy (14 layers)	$+0.535$ lift	S9-LA
uniform (28 layers)	$+0.530$ lift	S9-LA
U_shape (27 layers)	$+0.470$ lift	S9-LA
first10_only (10 layers)	$+0.460$ lift	S9-LA
odd_layers (14 layers)	$+0.345$ lift	S9-LA
V_shape (27 layers)	$+0.225$ lift	S9-LA
back_heavy (14 layers)	$+0.065$ lift	S9-LA
Multi-basis steering (S9-MB)		
MD-PCA-oppose $[-0.5, +0.5, 0]$	$+0.580$ lift	S9-MB
MD-only $[-0.5, 0, 0]$	$+0.525$ lift	S9-MB
MD+LR $[-0.3, 0, -0.2]$	$+0.520$ lift	S9-MB
Orthogonality (mean across layers 0, 10, 20)		
Brightness MD orthogonality	0.810 ± 0.060	FIX-ORTH
Warmth MD orthogonality	0.767 ± 0.076	FIX-ORTH
Colorfulness MD orthogonality	0.799 ± 0.045	FIX-ORTH
Animal MD orthogonality	0.730 ± 0.145	FIX-ORTH
Natural MD orthogonality	0.729 ± 0.144	FIX-ORTH
AdaLN survival ρ (mean across layers)		
Brightness	0.013	FIX-ADALN
Warmth	0.010	FIX-ADALN
Colorfulness	0.014	FIX-ADALN
Model comparison correlations		
Front-weighted model (r)	0.949	FIX-MODEL
Stability-alpha model (r)	0.933	FIX-MODEL
Pure stability model (r)	0.617	FIX-MODEL
Layer count model (r)	0.538	FIX-MODEL